ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 21-Sep-2026

Site: Site 1 Test Engineering | Location: Bengaluru, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
21.09 kW	**34,010 kWh**	**Rs. 976,516**	**4Y 7M**
21.45 kWp Installed	CUF 18.1% | PR 75%	Rs. 9.77 Lakhs	Profit: Rs. 100.25L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Site 1 Test Engineering
Location / Address	Bengaluru, Karnataka
Contact Email Address	test@example.com
Design Path Selected	Rooftop Area Analysis
System Target Capacity	21.09 kW
Total Installed Array Rating	21.45 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	39 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	2400.00 Sq.Feet (222.97 sq.m)
Utilized Usable Rooftop Area	2109 Sq.Feet (195.93 sq.m)
First Year Annual Generation	34,009.82 kWh
25-Year Cumulative Generation	755,625.55 kWh
Estimated Payback Period	4 Years & 7 Months
Post-Payback Profitability (25 Yrs)	Rs. 10,025,037.22 (INR 100.25 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 4,714,323.73 (INR 47.14 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 619 Tonnes CO2 | Tree Equivalence: 990 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Building 1 (Rooftop Array)
Gross Rooftop Area	2400.00 Sq.Feet (222.97 sq.m)
Utilized Effective Rooftop Area	2109 Sq.Feet (195.93 sq.m)
Building Plant Target Capacity	21.09 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	39 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Inverter Rating	20 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications

Core inverter rating, MPPT topology, and operating parameters

Selected Inverter Rating	20 kW Grid-Tied Solar Inverter
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx35 Sq.mm
DC Array Isolator Switch	80A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	40 Amps
Rated Inverter Operating Current	27.82 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx35 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx16 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	15.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550 W mono PERC / TOPCon	39 Nos.	Rs. 390,000
02	On-grid inverter	20 kW, three-phase	1 No.	Rs. 210,000
03	Rooftop mounting structure	HDGI rooftop structure, module-compatible	1 Set	Rs. 101,232
04	DC solar cable	4-6 sq.mm copper, PV-rated	231 m	Rs. 19,719
05	AC cable	4-10 sq.mm copper	126 m	Rs. 50,616
06	MC4 connectors	PV-rated connector pairs	39 pairs	Rs. 11,700
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 18,000
08	ACDB	40A MCB + 63A RCCB + Type-II AC SPD	1 No.	Rs. 18,000
09	Earthing kit	Chemical earth electrodes & accessories (IS 3043)	3 Nos.	Rs. 9,000
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 8,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 40,071
12	Installation & commissioning	Labour, testing and commissioning	1 Job	Rs. 100,178
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 976,515.65**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 5,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 5,000 (Refundable utility escrow)

STATUTORY COMPLIANCE & NET-METERING POLICY

All electrical installations conform to CEA (Technical Standards for Connectivity of Distributed Generation Resources) Regulations 2013 and State Grid Code. Net-metering bi-directional billing operates under applicable KERC / State SERC LT Solar Tariff orders.

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 976,516	**4Y 7M**	**Rs. 100.25L**	**Rs. 47.14L**
12-Item BOM Basis	Break-Even Year	Rs. 10,025,037	Alternative Asset

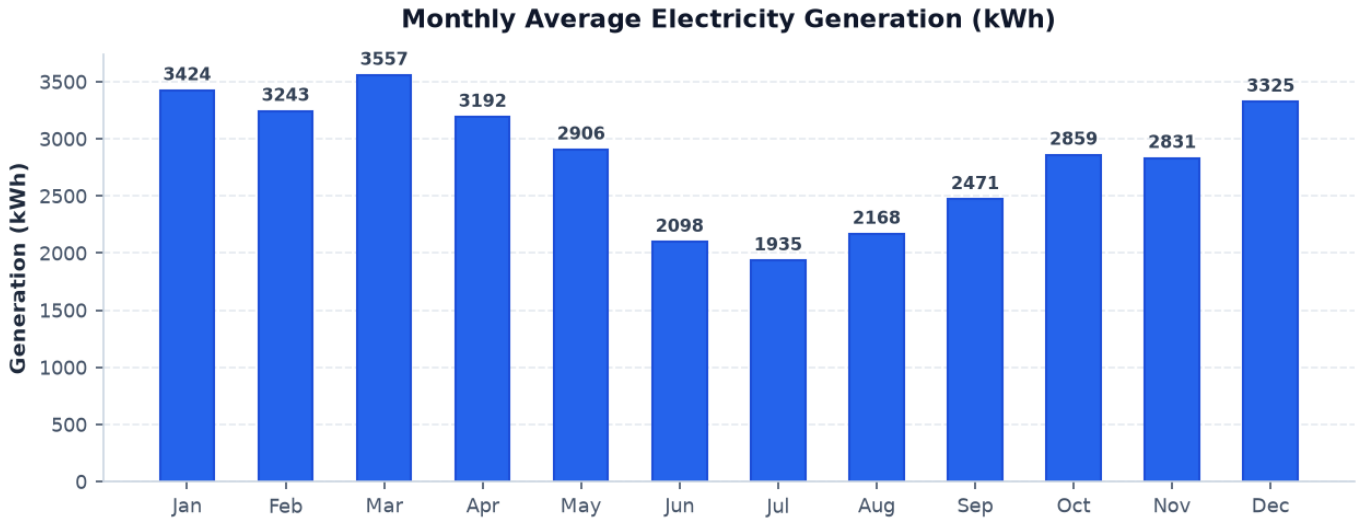
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	34,010	Rs. 8.05	273,779	12,654	261,125	261,125
02	33,670	Rs. 8.45	284,509	12,654	271,855	532,980
03	33,333	Rs. 8.87	295,664	12,654	283,010	815,990
04 *	**33,000**	**Rs. 9.32**	**307,557**	**12,654**	**294,903**	**1,110,893**
05	32,670	Rs. 9.78	319,510	12,654	306,856	1,417,749
06	32,343	Rs. 10.27	332,163	12,654	319,509	1,737,258
07	32,020	Rs. 10.78	345,171	12,654	332,517	2,069,775
08	31,699	Rs. 11.32	358,837	12,654	346,183	2,415,957
09	31,382	Rs. 11.89	373,137	12,654	360,483	2,776,440
10	31,069	Rs. 12.48	387,736	12,654	375,082	3,151,522
11	30,758	Rs. 13.11	403,236	12,654	390,582	3,542,103
12	30,450	Rs. 13.76	418,996	12,654	406,342	3,948,445
13	30,146	Rs. 14.45	435,607	12,654	422,953	4,371,398
14	29,844	Rs. 15.18	453,037	12,654	440,383	4,811,781
15	29,546	Rs. 15.93	470,666	12,654	458,012	5,269,793
16	29,250	Rs. 16.74	489,652	12,654	476,998	5,746,791
17	28,958	Rs. 17.57	508,791	12,654	496,137	6,242,928
18	28,668	Rs. 18.45	528,931	12,654	516,277	6,759,205
19	28,382	Rs. 19.37	549,753	12,654	537,099	7,296,304
20	28,098	Rs. 20.34	571,510	12,654	558,856	7,855,160
21	27,817	Rs. 21.36	594,168	12,654	581,514	8,436,674
22	27,539	Rs. 22.43	617,693	12,654	605,039	9,041,713
23	27,263	Rs. 23.55	642,051	12,654	629,397	9,671,110
24	26,991	Rs. 24.73	667,479	12,654	654,825	10,325,936
25	26,721	Rs. 25.96	693,671	12,654	681,017	11,006,953

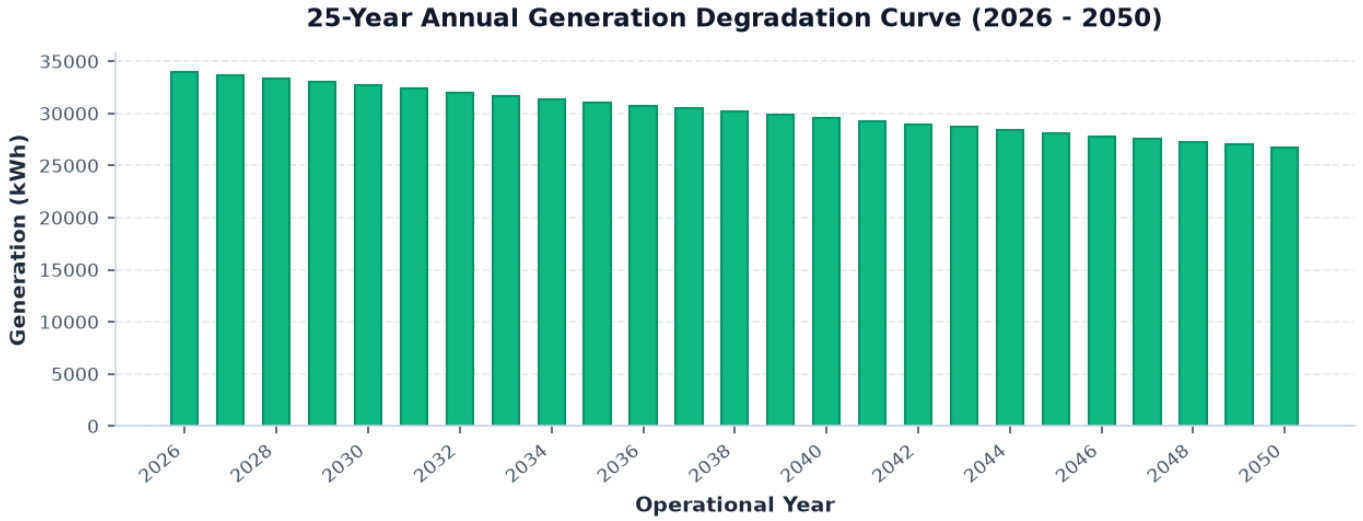
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD	PERFORMANCE RATIO	CAPACITY UTIL.	DEGRADATION RATE
YIELD	PR	CUF	DEGRADATION
1613 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

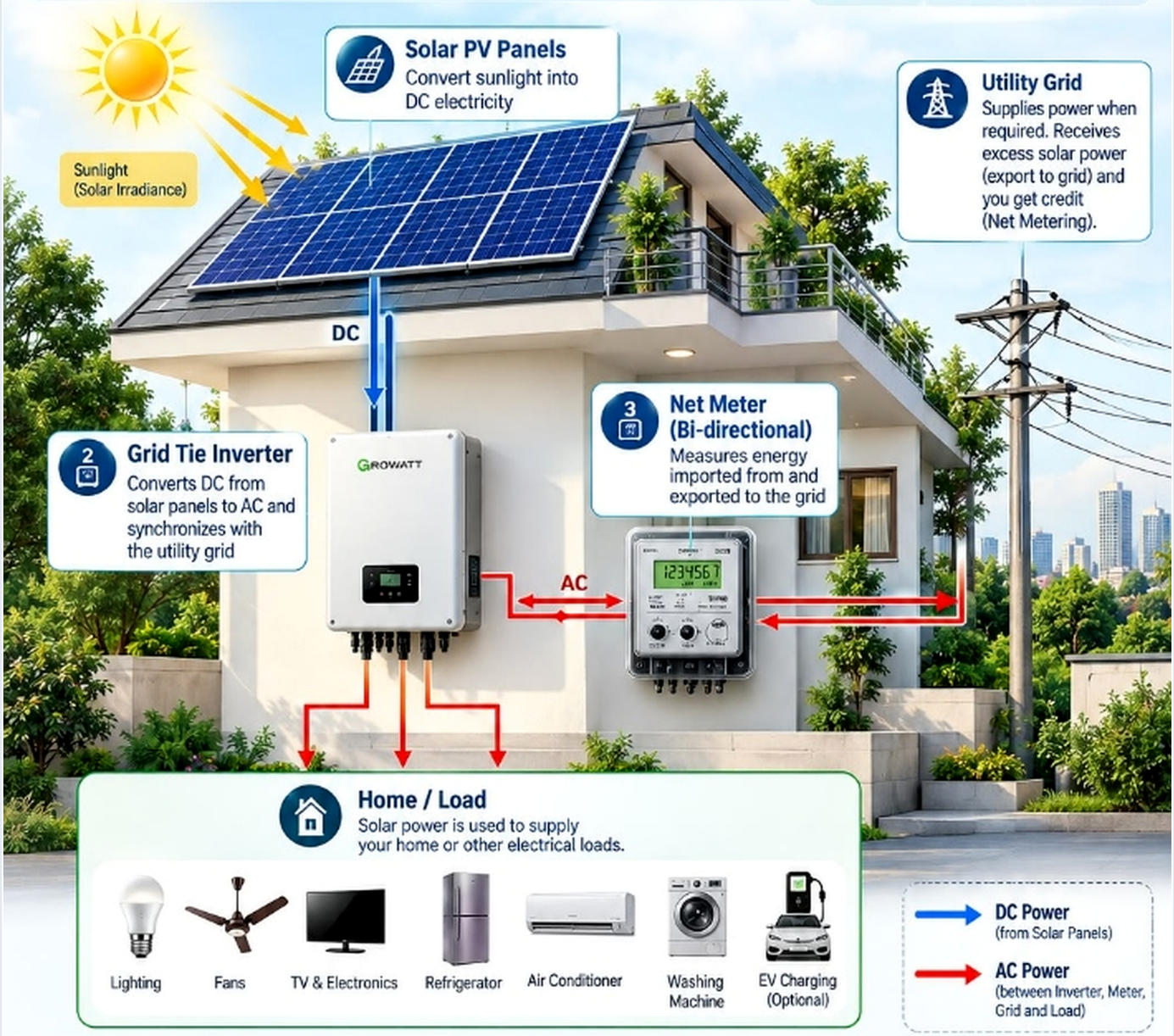
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow